

Reviewer 1

Comments

Reviewer 1 - Initial submission, 12/09/2026
 Major revision is required before submission.
Attached file: INS-D-26-9116 Review Report.docx

Reviewer 2

Comments

Reviewer 2 - Initial submission, 28/09/2026

The manuscript proposes Causal WeatherGraph, a regime-aware framework for identifying regional lagged dependencies in global atmospheric reanalysis data. It aggregates six-hourly ERA5 observations from 1979–2025 into 66 regional multivariate time series, constrains candidate edges using physical knowledge, and applies Granger-style regression across seven atmospheric regimes. The analysis combines FDR correction, randomized controls, temporal falsification, multivariable adjustment, PCMCI, and bootstrap analyses to investigate Wind–Humidity–Cloud pathways. The application is relevant, and the distinction between observational dependencies and interventional effects is welcome. However, several methodological issues require resolution before the central conclusions can be supported.

(1) The validity of significance testing under strong temporal dependence requires further assessment. Section 3.7 applies conventional OLS-based F-tests while controlling only three target-history lags in the principal model. Including these lags does not establish that residual autocorrelation and heteroskedasticity have been adequately addressed. Year-block bootstrap intervals for selected strong edges cannot establish calibration of the full discovery procedure. Please report residual diagnostics and reassess significance using inference procedures appropriate for serial dependence. Simulations with known dependency structures should evaluate the empirical false-discovery rate. Section 3.8 should also distinguish FDR control within each regime–target-variable group from any claim of control across the complete collection of 18,018 tests.

(2) The candidate graph and nearly saturated discoveries complicate the interpretation of pathway recovery and stability. Section 3.5 explicitly permits Wind → Humidity and Humidity → Cloud while excluding their reverse directions from the principal search space. Consequently, the orientation of the central pathway is partly imposed rather than discovered. Furthermore, Table 11 reports that 857 of 858 candidate edges are significant in at least one regime or lag. Under such saturation, high graph overlap across periods or candidate-neighborhood settings may provide limited evidence of structural specificity. The significant proportions for physical and distance-matched random candidates are also relatively close, at 0.792 and 0.743. Please evaluate bidirectional candidates under comparable spatial constraints and quantify pathway enrichment and graph overlap against repeated null graphs preserving relevant density, degree, and distance characteristics. The manuscript should clearly separate prior-imposed structure from information contributed by the observations.

(3) Regime definitions may introduce endogenous selection into the comparisons. Section 3.6 defines humidity and cloud regimes using contemporaneous spatial averages that include the response variable being analyzed. Differences between regime-specific graphs may therefore reflect response-dependent selection or truncation, in addition to

genuine changes in atmospheric coupling. Matching sample sizes addresses differences in statistical power but does not resolve this concern. Please include regime definitions based on appropriately preceding information or external circulation classifications, and directly test differences in source effects across regimes. In addition, Sections 4.3 and 5.1 interpret stronger global DJF relationships as stronger dependence under winter conditions. Because DJF corresponds to opposite seasons across hemispheres, this interpretation requires separate Northern and Southern Hemisphere analyses.

(4) Path composition and lag summaries do not yet establish a propagation or mediation mechanism. Section 3.9 constructs Wind–Humidity–Cloud chains by joining significant edges, but two predictive associations do not by themselves demonstrate transmission through the intermediate variable. The humidity-adjustment diagnostic uses humidity in the target region, whereas the enumerated chains can involve a different intermediate region. It therefore does not directly validate the cross-region pathways being counted. The mean coefficient attenuation of only 3.69% in the all-sample regime also warrants caution. Moreover, mean edge lags near two time steps and chain lags near four time steps are close to what approximately uniform detection over the tested one-to-three-lag range would produce. The shift toward mean lags around six when the maximum lag is extended to twelve raises a related search-window concern. Please evaluate specific intermediate regions with temporally consistent pathway models and assess lag-specific predictive contributions with uncertainty estimates. Physical transport interpretations would also benefit from wind-vector or moisture-flux diagnostics, since scalar wind speed does not encode transport direction.

(5) PCMCI confirmation is not independent validation, and the methodological contribution needs stronger benchmarking. Section 4.8 selects the strongest stable edges using the complete 1979–2025 record and then evaluates them with PCMCI on the overlapping 2019–2025 segment. The reported confirmation of 50 out of 60 edges demonstrates agreement between procedures on selected relationships, but does not constitute independent temporal validation or establish accuracy across the full graph. Please use an earlier period for discovery and parameter selection, reserving a later period exclusively for evaluation. Validation should include ordinary significant edges and unselected candidates, not only the strongest discoveries. The framework should also be compared with multivariable Granger, PCMCI, and existing regime-dependent approaches under comparable conditions. Known-structure simulations, component ablations, and runtime and memory measurements would help establish the specific benefits of the proposed framework.

Reviewer 3

Comments

Reviewer 3 - Initial submission, 22/09/2026

This manuscript proposes Causal WeatherGraph, which combines physics-constrained candidate graphs, regime-wise Granger-style testing, FDR control, and layered falsification on ERA5 data from 1979 to 2025. This manuscript is well-structured and its validation part is extensive, but several issues remain essential to the methodological contribution.

1 Regime-dependent causal discovery has already been studied by Regime-PCMCI, and recent work such as CaStLe addresses scalable space-time causal discovery through locality constraints in Earth-system data. The contribution here appears to lie mainly in combining physics-guided candidate restriction, manually defined atmospheric regimes,

large-scale ERA5 analysis, and a layered validation chain. The authors should explain more precisely what new causal discovery capability is added beyond these prior approaches and include direct methodological comparisons.

2 The true directed graph of the real ERA5 data is unavailable. A method paper needs evaluation where the causal structure is known. PCMRI is applied only to 60 preselected top stable edges from 2019 to 2025, with 50 confirmed, so this is a selective confirmation rather than a full method comparison. Synthetic or semi-synthetic spatio-temporal benchmarks with known direction, lag, and graph structure should be added, together with precision, recall, F1, SHD, or related graph-recovery measures and comparisons with relevant causal discovery baselines.

3 The main estimator uses OLS nested F-tests with three target-history lags (as defined in Section 3.7, Eqs. (21)-(25)). However, FDR correction does not correct p-values that are themselves distorted by residual serial dependence. The block-bootstrap analysis in Section 3.10 uses only 20 replicates for stability selection. Residual autocorrelation should be diagnosed, and the authors should consider block-bootstrap, HAC, or other dependence-aware inference for the main edge tests.

4 The regime construction can condition on the current outcome variable. High-humidity and high-cloud regimes are defined from contemporaneous humidity and cloud indices, and the regime mask is applied to the response time t . Thus, for relations such as $H_{\{t-l\}} \rightarrow C_t$, the high-cloud analysis conditions directly on a quantity derived from C_t . Matched sample sizes do not address this selection issue. Sensitivity analyses using lagged or otherwise externally defined regime indicators are needed.

5 Wind is reduced to scalar speed w , which contains no transport direction. However, cross-region $W \rightarrow H$ edges are interpreted in terms of moisture transport. The authors should either incorporate vector wind information or narrow this interpretation.

6 The falsification results raise a substantive directionality issue. In Table 8, forward $H \rightarrow C$ has a significant ratio of 0.740, whereas $C \rightarrow H$ reaches 0.801, and future-to-past Humidity to Cloud Cover also remains strong. It indicates that the direction of the humidity-cloud relation is not cleanly identified under the current controls. The main interpretation should reflect this more directly, and additional conditioning on common atmospheric drivers would be useful.

7 The very high discovery rate needs a stricter null calibration. The main analysis reports 14274 significant tests out of 18018, or 0.792. More importantly, the distance-matched random graph still has a significant ratio of 0.743 in Table 6. This small separation suggests that spatial dependence accounts for much of the raw significance. Together with the very small median partial R^2 values in Table 17, this study would benefit from an empirical-null comparison or an effect-size criterion in addition to $q < 0.05$.